

Natural Language Processing

Lecture 2: Levels of Representation and Ambiguity

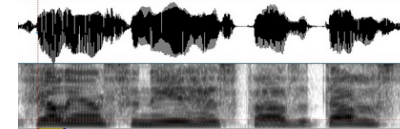
9/7/22

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Levels of Linguistic Representation

phonetics
phonology

sounds and sound
patterns of language



/bɔɪ/

morphology

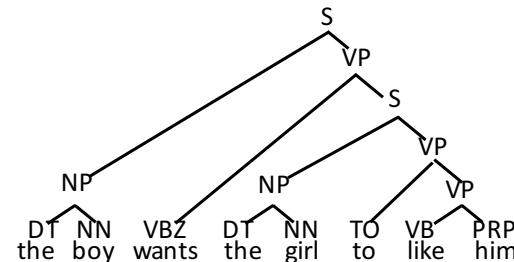
formation of words

in- + validate + -ed

DT	NN	VBZ	DT	NN	TO	VB	PRP	.
the	boy	want+s	the	girl	to	like	him	.

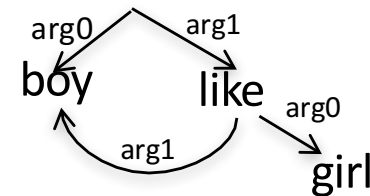
syntax

word order



semantics

word and sentence
meaning



pragmatics

influence of context and
situation

Language Understanding / Generation as Translation

- Most NLP techniques can be understood as translation tasks from one structure (level of representation) into another.
- For each translation step:
 - Construct search space of possible translations.
 - Find best paths through this space (decoding) according to some performance measure.
- Modern NLP relies on Machine Learning to figure out these translation steps.

NLP is hard: Ambiguity

- Unlike artificial languages, natural language is full of ambiguity.
- This can happen **on all levels of representation.**
 - *“Wreck a Nice Beach”* , *“Recognize Speech”*
 - *“inflammable”* = ***in*** + ***-flammable***
 - *“Enraged Cow Injures Farmer with Axe”*
 - *“Stolen Painting Found by Tree”*
 - *“Red Tape Holds Up New Bridges”*
 - *“Mouse”*



Lexical Ambiguity and Translation



Tobi Kellner

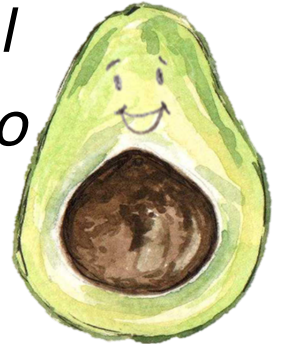
May 19 at 5:47 AM · 🌐

Can you tell what happened here?



Lost in Translation

fr: *L'avocat général* → en: *the attorney general*
→ en: *the general avocado*



en: *the spirit is strong but the flesh is weak*

⋮
↓
ru: ...
↓

en: *the vodka is good but the meat is rotten*

More Real Headlines

- *Ban on nude dancing on Governor's desk*
- *Eye drops off shelf*
- *Kids Make Nutritious Snacks*
- *Drunk gets nine months in violin case*
- *Government head seeks arms*
- *Patient at death's door – doctors pull him through*
- *In America a woman has a baby every 15 minutes*

Syntactic Structure

- What is the **part-of-speech** of each word? (noun, verb, adjective, adverb, determiner, ...)
- What are the **constituents**:
 - Noun phrase: *“Enraged cow”*, *“The cat with the hat”*, *“Columbia University”*
- What are the **subjects and objects**:
 - *“Dog bites man”* vs. *“Man bites dog”*
- **Modification**:
 - *“John saw the man in the park with a telescope”*

Structural Ambiguity

- Interplay between constituent structure and modification.
- Prepositional Phrase (PP) attachment:

Enraged cow injures farmer with axe.

Representing Modification with Brackets

[Enraged cow] [injures [farmer [with axe]]]
NP NP PP

[Enraged cow] injures [farmer] [with axe]
NP NP PP

More PP attachment

[Ban] on [rule banning] on [even's desk]

NP

NP

NP

- What are the possible modifications? Which one is correct?

Noun-Noun Modification

- Compound nouns also have internal structure:

country song platinum album

1. *[country [song [platinum album]]]*

2. *[country [[song platinum] album]]*

3. *[[country song] [platinum album]]*

4. *[[country [song platinum]] album]*

5. *[[[country song] platinum] album]*

Noun-Noun Modification

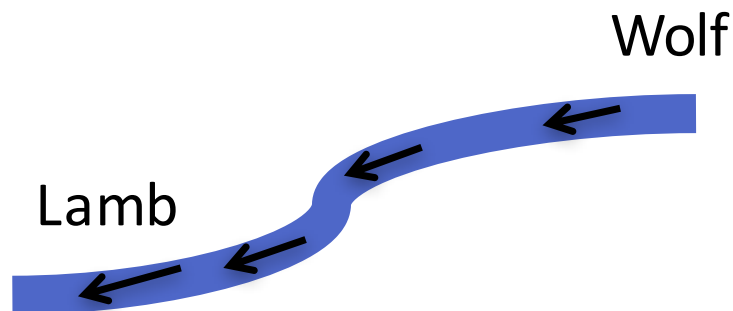
- What is the *semantic* relationship between nouns in a noun compound?
 - *Water fountain:* A fountain that **supplies** water.
 - *Water ballet:* A ballet that **takes place** in water.
 - *Water meter:* A device that **measures** water.
 - *Water barometer:* A barometer that **uses** water (instead of mercury) to measure air pressure.
 - *Water glass:* A glass that is meant to **hold** water.

Other tricky phenomena

- Need for deep semantic knowledge.

There was once a Wolf who saw a Lamb drinking at a river and wanted an excuse to eat it.

*For that purpose, **even though** he himself was **upstream**, he accused the Lamb of stirring up the water and keeping him from drinking. . .*



Other tricky issues:

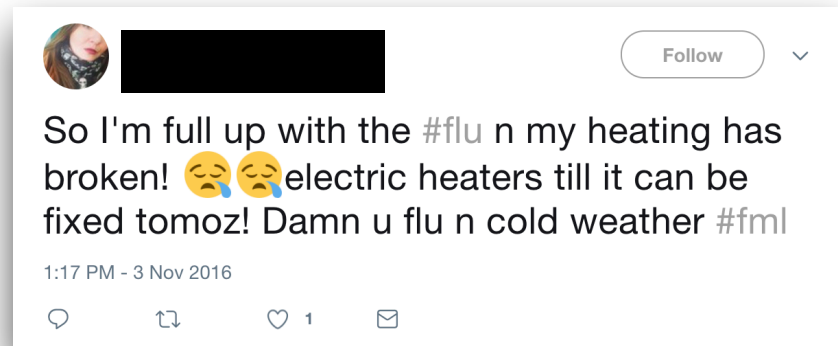
Language Variety

- Problem: Most NLP techniques were developed on English (specifically financial news written in American English in the 1990s), or other languages with many resources.
- Languages use different mechanisms to express meaning (morphology vs. word-order).



Other tricky issues: Domains and Language Change

- Non-standard English



- Idioms: *throw in the towel, get cold feet, kick the bucket*
- Neologisms (fixed lexicon doesn't work)
 - *noob, crowdsource, unfriend, retweet, bromance, ...*

Morphology

- Structure and formation of words.
- **Derivational** morphology: Create new words from old words (can also change the part-of-speech).

anti- + dis- + **establish** + -ment + -arian + -ism

- **Inflectional** morphology:
 - Convey information about number, person, tense, aspect, mood, voice, and the role a word plays in the sentence (case).
 - English has few morphological categories, but many languages are morphologically rich.

Morphology

- Morphological categories in English
 - Number (“*dog*”, “*dog +s*”)
 - Person (“*I run*”, “*She runs*”)
 - Tense (“*He wait+ed*”)
 - Voice (“*The issue **was** decid+ed*”)
- Other examples from other languages?